

Observation Recency in Bernoulli Fusion for Intermittent Multirobot Tracking

Abstract—Intermittent contact between mobile robots brings together target beliefs with different observation histories. Weighting an entire Bernoulli density by direct-observation age can improve existence decisions while degrading spatial pooling. We propose existence recency (ER), a fusion rule for labeled multi-Bernoulli tracking that screens untouched priors, retains ordinary spatial weights, and applies observation recency to existence terms. Both detections and missed detections update the local observation age, which remains unchanged by relaying. A two-block variational formulation establishes that this weighting preserves the fused spatial density for fixed inputs. Across three eight-robot simulation scenarios with twenty Monte Carlo trials each, ER reduces mean OSPA by 3.1% and 24.7% relative to its no-recency ablation in split-rejoin and churn-departure scenarios. Common-target localization RMSE remains within 0.02% of that ablation, whereas applying recency to both blocks increases it by 16.8% and 3.1%. ER yields higher post-departure false-target cost; adding two-hit confirmation reduces false hypotheses at the cost of slower discovery. These results identify a tradeoff between timely existence decisions and spatial information retention in multirobot fusion.

I. INTRODUCTION

Mobile robots often acquire complementary views while moving between connected and disconnected regions. When two subteams meet again, a newly received track can be supported by recent local measurements, inherited from an earlier encounter, or still close to an unobserved birth prior. Conversely, a low existence probability can reflect a useful recent missed detection or merely a lack of sensing opportunity. These distinctions matter when the communication opportunity permits only a small number of fusion rounds. The information exchanged over a newly available link can therefore differ substantially in recency. Figure 1 summarizes the proposed fusion mechanism.

Random finite set filters provide a natural representation of uncertain target number and state. The labeled multi-Bernoulli (LMB) filter represents each candidate track by an existence probability and a conditional spatial density [1]. Geometric pooling, also called generalized covariance intersection or Kullback–Leibler averaging (KLA), is widely used when cross-correlations are unavailable [2]. Unequal fields of view (FoVs) create differences in label support and observation history. We ask: *where should direct-observation recency enter a Bernoulli fusion rule when robots reunite?*

Multiplying a source’s entire fusion weight by an age factor seems natural, but changes both existence and position pooling. A source that has not observed a track recently can still carry a useful spatial estimate received from another robot. Reducing its spatial influence solely because its own sensor has been uninformative can therefore sacrifice

localization while improving target-number estimates. We study an existence-only alternative: retain ordinary spatial weights on eligible inputs, and use direct-opportunity age to weight the Bernoulli existence terms while retaining the spatial overlap normalizer.

We make three contributions. First, we introduce existence recency (ER): an observation-history rule qualifies fusion inputs, and the time since a robot’s last direct sensing opportunity determines its existence weight. Second, we derive the resulting Bernoulli pool from a two-block variational formulation and establish its fixed-input spatial invariance. Third, we evaluate discovery, departure, and false-track behavior in mobile robot teams, with ablations that separate input qualification, recency placement, and track confirmation.

II. RELATED WORK

A. Robot sensor fusion

Robot sensor fusion uses representations suited to the estimated state. The ROS robot_localization package provides generalized extended Kalman filter (EKF) fusion [3], while LIO-SAM organizes lidar and inertial constraints in a factor graph [4]. For dynamic-object tracking, Autoware combines data association with EKF state updates [5]. LMB representations also support robotic mapping: Gao *et al.* develop distributed multi-robot SLAM with LMB landmark maps and map fusion, evaluated on simulated and real data [6]. In our target-tracking setting, labeled Bernoullis represent object identity, existence, and state, while ER determines how robots combine their posteriors.

B. Cooperative robotic search and tracking

Dames studies distributed search and tracking with PHD filtering and robot motion [7]. Ramachandran *et al.* reconfigure a multirobot communication network after sensing-resource degradation, using a central optimization system [8]. DecSTER combines particle PHD filtering and Thompson-sampling action selection for decentralized search and tracking with unreliable, asynchronous inter-agent communication [9]. These works address decisions about where robots should move or how resources should be connected. We hold motion and attempted links fixed to isolate the effect of posterior fusion during intermittent contact.

C. Unequal views and information weighting

Wang *et al.* use information divergence to adapt LMB fusion weights across multiple views [10]. FoV compensation and multi-view geometric fusion are also established approaches [11], [12]. Gao *et al.* derive LMB-constrained

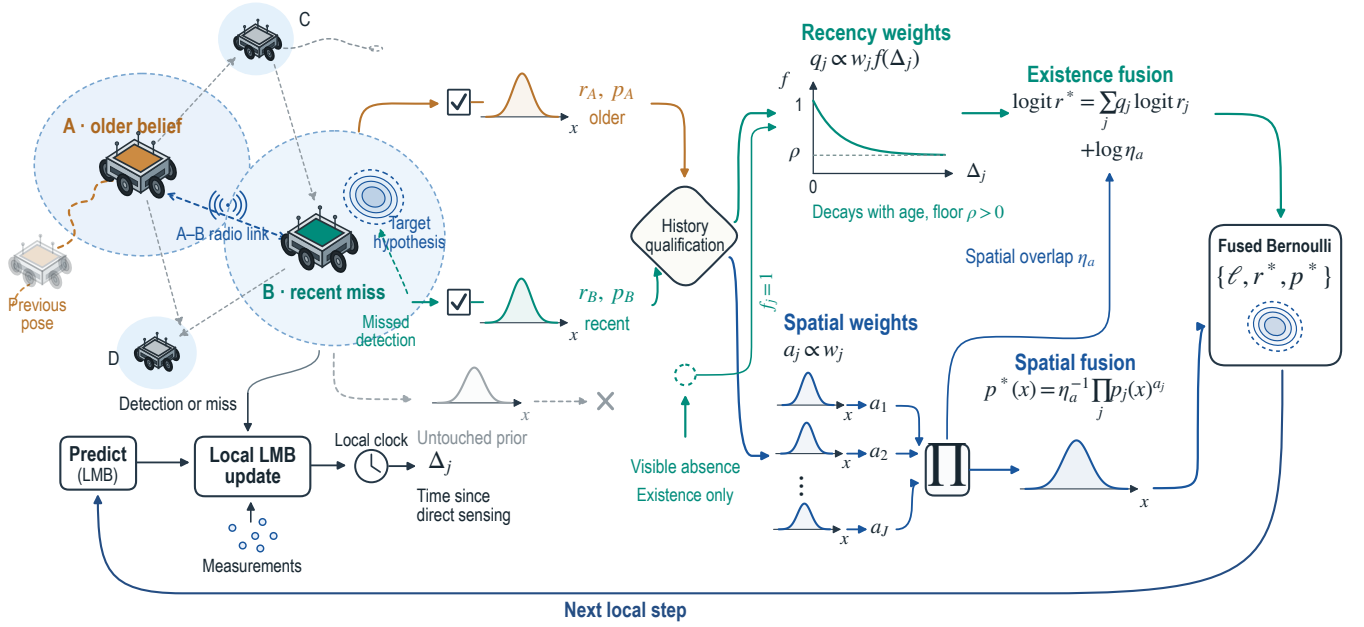


Fig. 1. Existence recency in multirobot tracking. Each robot predicts and updates an LMB, then exchanges posteriors at contact. Observation history qualifies fusion inputs; direct sensing age modulates existence weights q , while spatial pooling retains ordinary weights a . Visible absence contributes only existence evidence. The spatial-overlap normalizer η_a connects both branches, and the fused Bernoulli feeds the next local step.

minimum-information-loss (MIL) pooling and decompose label spaces for unequal views [13], [14]. Li *et al.* explicitly recognize that retained or relayed information can remain useful outside the current nominal FoV, and develop a density-decomposition CPHD approach [15]. Jin *et al.* examine shared-prior label consistency and introduce a consensus information selector [16]. Communication-aware adaptive KLA/LMB weights account for network constraints [17], while label assignment methods address inconsistent identities across nodes [18]. Building on these approaches, we examine where a direct-observation age factor should enter the Bernoulli fusion recursion.

D. Existence and spatial fusion

Üney *et al.* show that pointwise consistency of finite-set geometric mixtures does not ensure cardinality consistency, and advocate separate cardinality and localization fusion [19]. Our rule retains the spatial overlap penalty while assigning distinct weights to spatial densities and Bernoulli existence terms. This formulation lets us isolate the effect of observation recency on existence decisions and examine the resulting discovery–localization tradeoff at robot encounters.

III. PROBLEM AND FUSION RULE

A. Robot, sensing, and communication model

There are N robots with known poses $s_i(t)$ and a common clock. A physical edge exists when two robots are within radio range. A reliable geometry coordinator selects a bidirectional minimum spanning tree in each connected component. At each time, a robot predicts its LMB, updates with its own measurement set, and exchanges the resulting posterior for one synchronous fusion round. A dropped

neighbor’s Metropolis weight returns to the receiver’s self weight. Fusion output is retained as the prior for the next local step. Communication cost includes coordinator traffic and lossy posterior messages.

A label $\ell = (t_b, b)$ specifies a shared birth time and candidate region. For represented label ℓ , source j supplies $\pi_j^\ell = (r_j^\ell, p_j^\ell)$, where $r_j^\ell \in [0, 1]$ and $p_j^\ell(x)$ is a Gaussian-mixture density over position and velocity. All robots use the same birth priors and label keys.

B. Observation history and participation

Each retained Bernoulli carries two metadata values. The history flag h_j^ℓ records whether the density descends from any scheduled local update with nonzero detection opportunity. A direct opportunity is evaluated on the predicted Gaussian means before the measurement update:

$$o_j^\ell(t) = \mathbf{1} \left[\sum_g w_{jg}^\ell p_{D,j}(\mu_{jg}^\ell, t) > 0 \right]. \quad (1)$$

The local timestamp τ_j^ℓ is the last time this condition held at robot j . Both detections and missed detections count. A received posterior propagates the history flag, while the receiver retains its own local timestamp after fusion. The resulting age measures time since the last *direct sensor opportunity*.

For each label, if at least one input carries observation history, represented inputs with no such history are excluded as untouched priors. Otherwise they retain ordinary participation. An absent label contributes a negative existence sensor only when the represented spatial density places at least half its Gaussian-mean mass inside that source’s current FoV, with expected detection at least 0.1. Its existence upper

bound is 0.01. Other absent sources abstain. Such sensors are qualified negative evidence and are never assigned a spatial density. The flags propagate only through actual participating inputs. FoV occupancy is approximated at the Gaussian-component means.

Let $\mathcal{P}$ denote eligible represented inputs and $\mathcal{E}$ eligible existence inputs, including qualified absence sensors. For the receiver's delivered Metropolis weights w_j , define

$$a_j = \frac{w_j \mathbf{1}[j \in \mathcal{P}]}{\sum_{k \in \mathcal{P}} w_k}, \quad q_j = \frac{w_j f_j \mathbf{1}[j \in \mathcal{E}]}{\sum_{k \in \mathcal{E}} w_k f_k}. \quad (2)$$

For a represented label with a positive timestamp set

$$f_j = \rho + (1 - \rho) \exp[-(t - \tau_j^\ell) \Delta t / \lambda]. \quad (3)$$

Use $f_j = \rho$ for a represented but never-directly-observed label, and $f_j = 1$ for a qualified absence sensor. We set $\rho = 0.25$ and $\lambda = 5$ s. The age factor is independent of the sign of the existence claim: a recent low r_j can outweigh an older high r_j . Positive self weight and the existing absence logic supply the ordinary single-input fallback; a label without eligible spatial support is omitted.

C. Existence recency with fixed spatial weights

For clarity suppress the label and receiver indices. Denote Bernoulli relative entropy by D_B and ordinary density relative entropy by D . For fixed eligibility and normalized a, q , consider

$$\min_{r \in (0,1), p} J(r, p) = \sum_{j \in \mathcal{E}} q_j D_B(r \| r_j) + r \sum_{j \in \mathcal{P}} a_j D(p \| p_j). \quad (4)$$

When the two supports and weight vectors agree, this reduces to the Bernoulli KLA objective.

Proposition 1 (fixed-input form). Assume $0 < r_j < 1$, $\eta_a = \int \prod_{j \in \mathcal{P}} p_j(x)^{a_j} dx > 0$, and finite relevant divergences. A minimizer of (4) is

$$p^*(x) = \eta_a^{-1} \prod_{j \in \mathcal{P}} p_j(x)^{a_j}, \quad (5)$$

$$\text{logit } r^* = \sum_{j \in \mathcal{E}} q_j \text{logit } r_j + \log \eta_a. \quad (6)$$

For fixed a , eligibility, and input densities, changing q changes existence without changing p^* .

Proof. Writing $p_a = \eta_a^{-1} \prod_j p_j^{a_j}$ gives $\sum_j a_j D(p \| p_j) = D(p \| p_a) - \log \eta_a$. For every $r > 0$ this is minimized at $p = p_a$. Differentiating the remaining scalar objective yields $\text{logit } r - \sum_j q_j \text{logit } r_j - \log \eta_a = 0$. Its second derivative is $1/[r(1-r)] > 0$, establishing the stated solution. $\square$

For normalized densities, $0 < \eta_a \leq 1$, so the spatial overlap term remains a penalty in the existence update. Across recursive filtering steps, existence decisions can affect associations, pruning, and state extraction. We therefore evaluate localization on common assignment support alongside global set metrics.

The implementation calls the existing mixture-aware Gaussian-power fusion routine with a for spatial pooling and q for existence. It retains at most three components per input for the powered-mixture calculation, at most 256

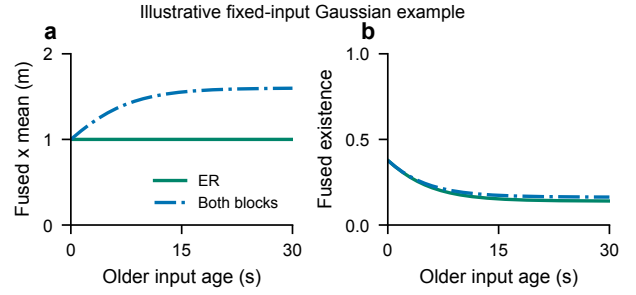


Fig. 2. Effect of observation age in a two-source Gaussian example. Two unit-covariance Gaussian densities have x means 0 and 2 m, with existence 0.9 at the older source and 0.1 at the current source. Ordinary spatial weights are equal. Increasing the older source's direct age changes ER's existence while leaving its spatial mean at 1 m. Applying age to both blocks also moves the spatial mean. Both rules retain their respective spatial overlap normalizer.

component tuples, and at most eight output components. A non-admissible approximate overlap normalizer triggers a moment-matched fallback. This provides a finite-mixture approximation of the weighted density product. All methods retain the same eight-component cap, prediction, measurement-update code, and MAP-cardinality extraction. We call the rule *existence recency* (ER); using q in both blocks is the distinct *ER (both)* ablation.

D. Confirmation ablation and communication cost

A separate ablation delays history-based input restriction until some input carries a positive confirmation. A robot creates this bit after two consecutive local updates with opportunity and detection-association mass at least 0.5. Empty measurements reset the local streak. The confirmation bit propagates through fusion, while the hit counter and observation timestamp remain local. Ordinary FoV-aware pooling applies before confirmation. This two-hit rule tests the tradeoff between false-track rejection and discovery during short sensing contacts.

All communicating methods transmit the full retained mixture and identical metadata. Each attempted packet occupies 16,384 bytes, with a fixed 128-byte-per-robot control charge per frame. Padding isolates fusion at equal attempted wire cost. We also report unpadded payloads because differing mixture sizes can change actual data volume. The method adds constant metadata per represented label and linear-time weight calculations.

IV. EXPERIMENTS

A. Mobile sensing scenarios

Eight simulated robots follow two smooth patrol families for 120 frames at $\Delta t = 0.5$ s. Their maximum speed is below 3 m/s and acceleration below 2.1 m/s². Radio range is 24 m and circular sensing range is 14 m. The fixed paths induce connected and two-component periods (Fig. 3). All targets use a constant-velocity state with independent per-step acceleration standard deviation 0.016 m/s². The filter uses a continuous white-acceleration covariance with

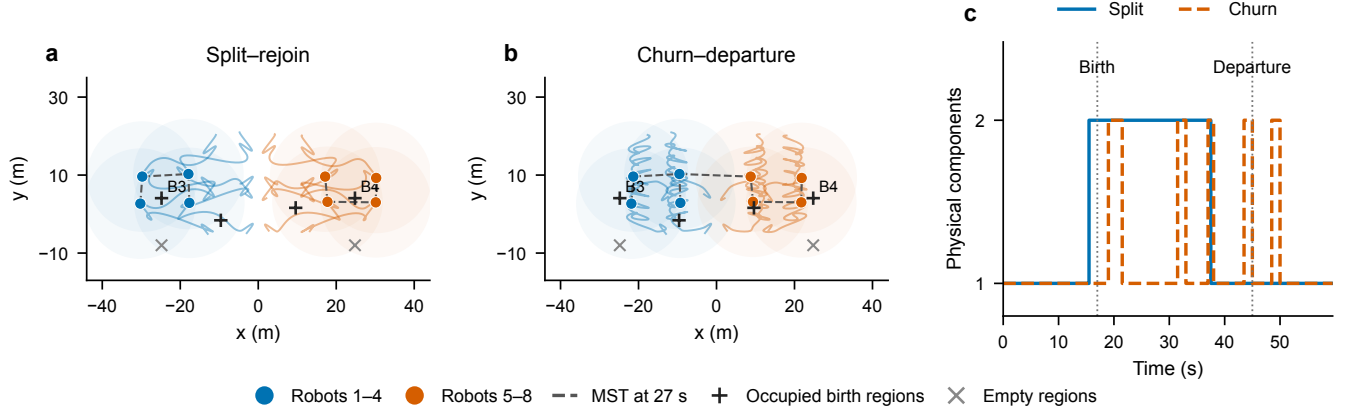


Fig. 3. Mobile sensing scenarios. Thin curves are robot paths; circles and dashed tree edges show the state at 27 s. Light disks denote 14 m FoVs. Crosses mark candidate birth regions; the two lower regions remain empty. The split and churn scenarios create different contact histories, with target departure at 45 s in the churn scenario.

intensity 0.0128 and survival probability 0.995. Within FoV, detection probability is 0.9 and Cartesian observation noise has standard deviation 0.4 m. Poisson clutter has mean one per robot per frame and is uniform over a fixed 104×80 m rectangle. Local pruning discards existence at or below 0.01; belief propagation uses tolerance 10^{-6} and at most 1,000 iterations. All methods share these settings.

The *split-rejoin* scene has two initial targets and two targets born at frame 35 while the teams are separated. The *churn-departure* scene uses repeated contact changes and removes one late target after frame 90; the filters receive no departure notification. In the *no-new* control, only the two initial targets exist, although identical later birth priors are still offered. Six fixed candidate regions have prior existence 0.1, position standard deviation 3.2 m, and velocity standard deviation 0.2 m/s. Regions 1–2 are activated at frame 1 and regions 3–6 at frame 35. Regions 5–6 remain empty in every scene. The filters receive the candidate birth schedule and priors.

Each scenario uses twenty Monte Carlo trials. In each trial, every robot receives an independent constant path offset drawn uniformly from $[-0.6, 0.6]^2$ m, and the communication components and trees are recomputed. The offsets vary the layout and contacts while preserving speed and acceleration. Target trajectories, detections, clutter, and directed link-loss draws are paired across methods. Directed packet loss is 0.1.

B. Controls and ablations

Local runs without communication. *FoV* uses FoV-based absence sensors. *ER w/o age* adds observation-history qualification and omits the age factor. *Age-all* applies direct-age weights to both spatial and existence pooling without history-based qualification. *ER (both)* adds this qualification, separating placement of the age factor from input selection. *ER* uses the same qualification with age in existence alone. *Confirmed ER* adds the two-hit gate.

The *MIL-Z* control uses constrained LMB MIL, $r =$

$\sum_j w_j r_j$ and $p = \sum_j w_j r_j p_j / r$, with zero existence for absent labels [13]. *MIL-S* specializes label-subspace MIL [13], [14] to the shared-label setting: for each subspace defined by a source-membership pattern, weights are renormalized over represented sources before applying the same pooling formula. Both MIL controls use the shared birth keys for label alignment and retain untouched represented priors. Their Gaussian-mixture unions are reduced to the common component cap.

Every communicating method uses the same attempted links, delivered links, weights before label processing, packet size, and control charge. The local update and extraction are also shared.

C. Metrics and uncertainty

We report position OSPA with order 2 and cutoff 12 m [20], cardinality absolute error, and GOSPA with $p = 2, c = 12$ m, $\alpha = 2$ [21]. Localization, missed-target, and false-target GOSPA components are reported as *squared costs* in m^2 . All extant targets are scored at every robot, including during disconnection. The scores therefore measure team-wide information coverage as well as local tracking.

For localization comparisons, assignments are computed separately for each method and matched errors are pooled only on robot–time–truth triples matched within the cutoff by both methods. We report the resulting RMSE and common-support size; unmatched targets contribute to OSPA/GOSPA. Departure false cost uses frames 91–120. Reunion OSPA uses the next ten frames, truncated at a further disconnection. For opposite-subteam robots, acquisition is defined geometrically as the first of three consecutive assignments within 2 m of a late target, censored at that target’s departure or the trial end.

We report paired mean differences and 95% percentile bootstrap intervals from 10,000 resamples of the twenty paired trials. Resampling whole trials retains the temporal and inter-robot dependence within each realization.

TABLE I

SET ESTIMATION OVER TWENTY PAIRED MONTE CARLO TRIALS PER SCENARIO. OSPA IS MEAN $\pm$ TRIAL SD (M); COUNT DENOTES MEAN ABSOLUTE CARDINALITY ERROR.

Method	Split-rejoin		Churn-departure		No-new control	
	OSPA	Count	OSPA	Count	OSPA	Count
Local	7.099 ± 0.454	1.368	7.280 ± 0.537	1.308	6.004 ± 0.839	0.645
FoV	3.713 ± 0.584	0.614	2.867 ± 0.551	0.442	0.804 ± 0.163	0.056
ER w/o age	2.651 ± 0.106	0.411	1.123 ± 0.421	0.143	0.797 ± 0.264	0.056
MIL-Z	4.812 ± 0.499	0.641	4.204 ± 0.574	0.452	2.765 ± 0.854	0.032
MIL-S	4.762 ± 0.506	0.624	4.112 ± 0.582	0.431	2.769 ± 0.854	0.033
Age-all	2.912 ± 0.123	0.466	1.960 ± 0.533	0.292	0.745 ± 0.172	0.043
ER (both)	2.589 ± 0.090	0.399	0.937 ± 0.250	0.115	0.752 ± 0.233	0.045
ER	2.569 ± 0.110	0.398	0.846 ± 0.205	0.103	0.668 ± 0.219	0.040
Confirmed ER	2.607 ± 0.107	0.403	1.519 ± 0.440	0.220	0.652 ± 0.157	0.037

V. RESULTS

A. Existence recency and set estimation

Table I compares set estimation across the three scenarios. Relative to ER without recency, ER reduces mean OSPA from 2.651 to 2.569 m in split-rejoin and from 1.123 to 0.846 m in churn-departure. The paired differences are -0.082 m $[-0.110, -0.051]$ and -0.277 m $[-0.399, -0.159]$, respectively. In the no-new control, the difference is -0.129 m $[-0.167, -0.095]$. Brackets denote the 95% bootstrap intervals defined above. Cardinality error decreases in all three scenarios. In the reunion windows, OSPA is 2.533 versus 3.264 m in split-rejoin and 0.926 versus 1.248 m in churn-departure (ER versus ER w/o age).

FoV and the two MIL controls have higher OSPA than ER. Support partitioning improves MIL in the two event scenarios, while both controls incur larger localization and false-target costs. Local has the largest OSPA, consistent with evaluating global target sets while robots are separated.

B. Where the age factor belongs

Table II and Fig. 4(b) distinguish existence gains from localization. With the same observation-history qualification, ER (both) increases common-target RMSE relative to ER w/o age by 16.8% in split-rejoin and 3.1% in churn-departure. ER leaves this RMSE almost unchanged, with ratios of 0.9999 and 0.9998. This agrees with the motivation in Fig. 2: recent direct sensor opportunity is useful for existence weighting, while a posterior also retains spatial information from earlier observations and robot encounters.

C. False hypotheses and the confirmation tradeoff

Table III shows the false-hypothesis tradeoff. ER's mean post-departure false cost is 12.255 m^2 , compared with 10.785 m^2 for ER w/o age. The paired difference is $+1.470 \text{ m}^2$ $[-0.705, 3.795]$. Similarly, ER's no-new false cost is 0.570 m^2 , versus 0.00375 m^2 for FoV. Lower OSPA therefore does not imply better false-track rejection.

Confirmed ER reduces no-new false cost to 0.00375 m^2 , but its churn OSPA is 1.519 m, versus ER's 0.846 m. This is consistent with the cost of requiring consecutive local positive associations during short sensing opportunities. In churn,

TABLE II

COMMON-TARGET LOCALIZATION RELATIVE TO ER W/O AGE. EACH RATIO DIVIDES THE POOLED METHOD RMSE BY THE REFERENCE RMSE; BRACKETS GIVE 95% BOOTSTRAP INTERVALS. n_c COUNTS COMMON ROBOT-TIME-TRUTH TRIPLES.

Scene	Method	RMSE ratio [95% interval]	n_c
Split	ER	0.9999 [0.9997, 1.0001]	57,988
Split	ER (both)	1.1685 [1.1270, 1.2087]	58,006
Split	Confirmed	1.0018 [1.0004, 1.0032]	57,894
Churn	ER	0.9998 [0.9997, 1.0000]	58,936
Churn	ER (both)	1.0307 [1.0218, 1.0390]	58,841
Churn	Confirmed	1.0028 [0.9937, 1.0124]	56,641

TABLE III

FALSE-TARGET GOSPA SQUARED COST (m^2), AVERAGED OVER TWENTY TRIALS. DEPARTURE USES FRAMES 91–120; NO-NEW USES THE FULL TRIAL.

Method	Departure	No-new
Local	10.935	0.611
FoV	6.675	0.004
ER w/o age	10.785	0.679
MIL-Z	23.430	4.162
MIL-S	26.370	4.204
Age-all	9.690	0.004
ER (both)	10.905	0.570
ER	12.255	0.570
Confirmed ER	10.815	0.004

ER acquires 160/160 opposite-subteam robot-target queries, while Confirmed ER acquires 152/160. Among successful acquisitions, mean delays are 2.47 and 7.63 s, respectively. The remaining eight Confirmed ER queries are unacquired by their censoring time. The unsmoothed trajectories in Fig. 5 show why discovery and post-departure persistence should be evaluated separately.

D. Communication cost

All communicating methods use the same attempted and delivered messages for each paired trial. Mean attempted wire cost is 24.98 MiB for split-rejoin and no-new, and 25.74 MiB for churn-departure. Across communicating methods

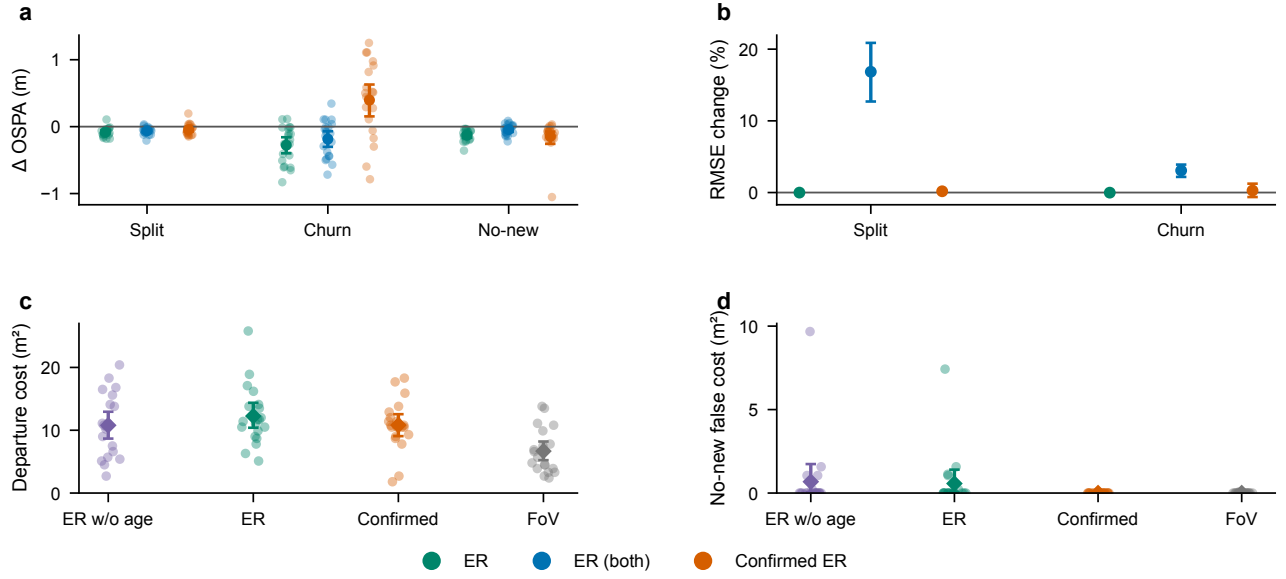


Fig. 4. Set error, localization, and false-target costs. (a) Paired OSPA differences relative to ER w/o age; small dots show individual trials, and large points with bars show means and 95% bootstrap intervals. (b) Percentage changes in pooled RMSE on common assignment support. (c,d) Departure and no-new false-target squared costs, with twenty trial points per method. ER reduces set error with little change in common-target localization; confirmation changes the discovery–false-cost tradeoff.

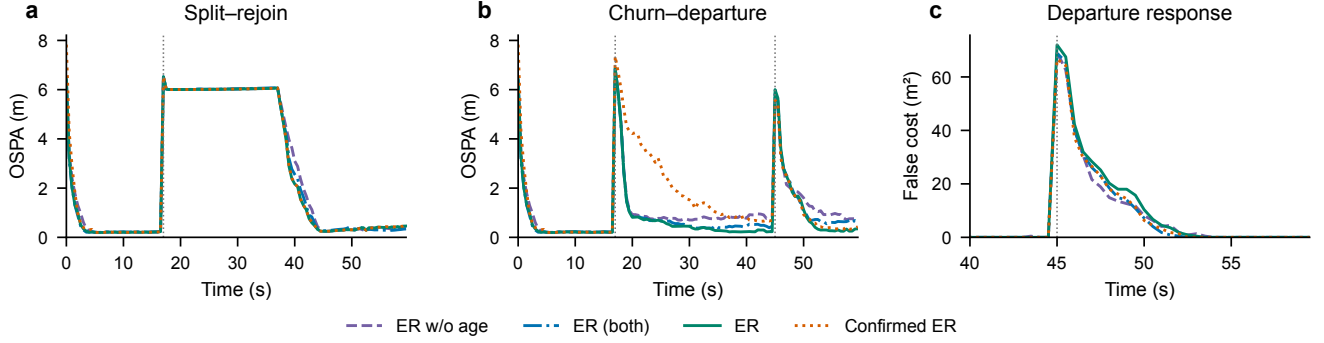


Fig. 5. Mean trajectories across twenty Monte Carlo trials; no smoothing is applied. Dotted event markers indicate late birth at 17 s and, in churn, departure at 45 s. ER and ER (both) improve set error relative to ER w/o age while differing in localization and false-target persistence.

and scenarios, mean unpadding payload totals range from 2.14 to 9.85 MiB; the padded message budget is identical.

VI. DISCUSSION AND CONCLUSION

ER uses observation history to select participating inputs and direct sensing age to weight existence, while retaining ordinary spatial weights. Across the three mobile-robot scenarios, this separation reduces set error relative to ER without recency. In split–rejoin and churn–departure, it also avoids the common-target localization degradation of applying age to both blocks. A robot’s direct sensing age and the spatial information accumulated in its posterior describe different properties: earlier observations and relayed evidence can remain useful after direct contact ends.

False-track rejection introduces a further tradeoff. ER retains some false hypotheses, and the two-hit confirmation gate reduces no-new false cost at the expense of discovery

during short sensing contacts. The departure and acquisition results show why set error, localization, false cost, and discovery delay should be evaluated together. An adaptive confirmation rule that accounts for sensing-contact duration is a concrete direction for extending this approach.

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